

Serie 4

LINEAR SYSTEMS, ROW REDUCTION

The first exercise is repeated from the previous exercise sheet, since the necessary material wasn't yet covered.

1. Linear system. Fix $b_1, b_2, b_3 \in \mathbb{R}$. Determine when the following linear system of equations has a solution and describe its set of solutions $S \subseteq \mathbb{R}^3$ when it does

$$\begin{pmatrix} 0 & 7 & -2 \\ -1 & 2 & -1/2 \\ 4 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

2. Linear system. For which values of $a \in \mathbb{R}$ and $b \in \mathbb{R}$ does the following system admit solutions $(x, y, z) \in \mathbb{R}^3$?

$$\begin{aligned} x + 2y - 3z &= 10 \\ 3x + 7y - 9z &= b \\ -3x - 5y + az &= b \\ 2x + 4y - 3z &= 5 \end{aligned}$$

For all such values of a and b give all the solutions.

3. Linear system.

Consider the following two systems of equations.

$$\begin{cases} x + y + z &= 6 \\ x + 2y + 2z &= 11 \\ 2x + 3y - 4z &= 3 \end{cases}$$

and

$$\begin{cases} x + y + z &= 7 \\ x + 2y + 2z &= 10 \\ 2x + 3y - 4z &= 3 \end{cases}$$

Solve both systems of equations simultaneously using row-reduction to an appropriate 3×5 extended matrix.

4. Row reduction. You learnt in class that if the augmented matrices of two linear systems of equations are equivalent, they have the same solutions. Is the converse true? If yes, prove it, and if not, provide a counterexample.

5. Row reduction. Let A be a 2-times-2 real matrix and let $b \in \mathbb{R}^2$ such that the system of equations $Ax = b$ has no solutions. Show that one row of the matrix A is a multiple of the other.